

Week 3 – Docker Containerization

Summary:

This week focused on **containerizing the backend using Docker** to ensure portability and isolated execution environments. All compiler services were integrated into Docker containers with pre-installed libraries and compilers.

Key Activities:

- Studied Docker concepts and container architecture.
- Dockerized FastAPI backend.
- Installed compilers (GCC, g++, JDK) and Python libraries inside the container.
- Configured **volume mapping** for code execution and result retrieval.
- Tested all endpoints inside the Docker container.

Technologies Used: Docker, WSL2, FastAPI, Python, GCC, g++, JDK

Outcome:

Backend is fully functional inside a Docker container, providing a stable and reproducible environment for multi-language execution.